

AUSTIN LEVERS OF ECONOMIC MOBILITY INDEX (LEMI)

TECHNICAL REPORT

Background, Methodology, and Analysis

April 2026

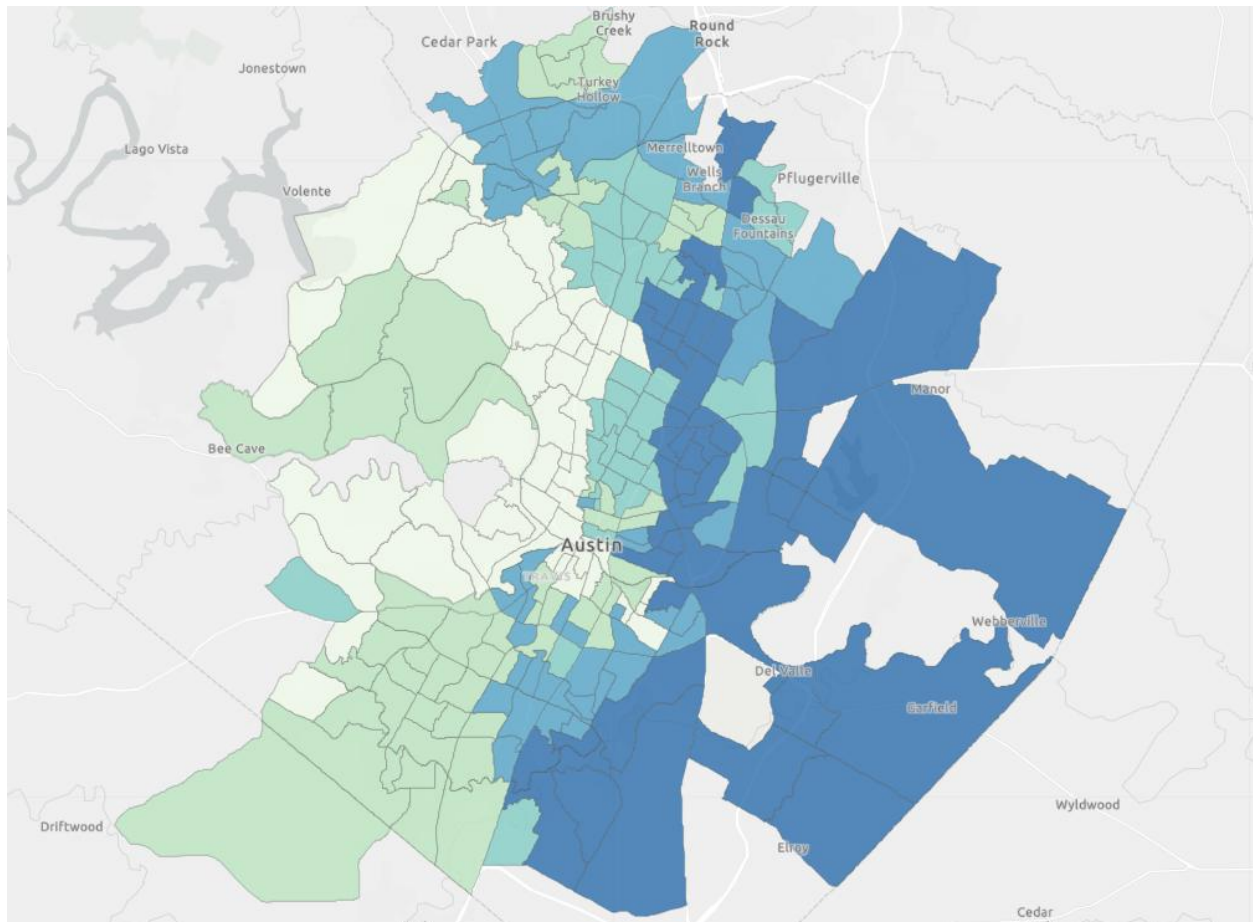


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INTRODUCTION

The Austin Levers of Economic Mobility Index (LEMI) is a tract-level composite measure developed to assess how neighborhood conditions support or constrain economic mobility. Rather than evaluating individual outcomes, the Index examines the broader structural environment in which residents live, work, and access services. It is intended to capture how place influences opportunity over time.

LEMI functions as an analytic framework for comparing neighborhood conditions across the city. It provides a standardized way to identify geographic variation, surface spatial patterns, and support strategic planning decisions. While the [StoryMap](#) serves as the primary public-facing platform for exploring results, this technical report documents the theoretical foundation, data sources, methodological approach, and scoring procedures used in constructing the Index.

The Index is structured around three core domains: Health & Wellbeing, Livelihood & Work, and Access & Belonging. These domains reflect the understanding that economic mobility is shaped by interconnected systems, including health stability, employment opportunity, and the ability to access and participate in civic and community life. The methodology draws on established composite indicator practices, including guidance from the Organisation for Economic Co-operation and Development (OECD) Handbook on Constructing Composite Indicators, and the Esri Technical Paper on Creating Composite Indices Best Practices with an emphasis on clarity, reproducibility, and balanced representation across domains. The final dataset includes 249 census tracts within the study area.

BACKGROUND

Predecessors and Local Tool Landscape

The LEMI was developed in a policy environment where the City already uses a variety of dashboards, sector-specific prioritization tools, and screening frameworks. These tools provide valuable domain-level information but are not designed to offer a unified, cross-sector view of structural economic mobility conditions across neighborhoods.

The LEMI fills a distinct role by providing a regionally consistent, cross-domain composite measure of economic mobility-related conditions. It is intended to complement, not

replace, existing tools by offering a common geographic baseline to align priorities, coordinate cross-department efforts, and understand program footprints.

To ensure the LEMI complements existing City efforts rather than duplicating them, the Equity and Inclusion team conducted a structured review of current dashboards, scoring tools, spatial analyses, and planning frameworks used across departments. This review examined each tool's purpose, strengths, methodological approach, and relationship to LEMI's regional, structurally focused composite index.

Following review, existing City tools primarily fall into three categories: sector-specific prioritization tools, operational dashboards, and qualitative screening frameworks. LEMI fills a distinct role as a region-wide, cross-sector composite index designed to measure structural conditions consistently across neighborhoods.

LITERATURE REVIEW

Composite indices and opportunity/equity mapping have become widely used in U.S. cities and at the federal level as a way to synthesize complex, multidimensional conditions into interpretable neighborhood geographies. This tradition includes early “opportunity mapping” approaches used to support fair housing strategies and avoid concentrating assisted housing in high-poverty areas, as well as contemporary composite indices used for capital planning, climate investment, and public health resource targeting.

The OECD composite indicator framework codifies a common workflow: defining a conceptual model, selecting indicators, preprocessing and normalizing data, choosing a weighting and aggregation method, and conducting validation and sensitivity testing. The LEMI follows this workflow, with design choices made to prioritize interpretability and long-term maintainability in a municipal context.

Methodological Options: Weighting, Normalization, and Interpretability

Constructing a composite index requires navigating several unresolved methodological questions, with weighting standing out as one of the most debated elements. While practitioners often rely on expert judgment, statistical techniques like Principal Component Analysis (PCA), or pragmatic constraints, there is rarely a single correct weighting scheme. For decision-support tools, equal weighting is frequently adopted; this approach is most effective when the conceptual framework treats all domains as inherently equal and the primary goal is to provide a balanced diagnosis rather than a specific prediction.

Normalization represents the second major design option, where percentile ranking has become a staple for public-facing indices. This method is favored because it is robust against outliers, easy for non-experts to interpret, and successfully aligns indicators measured in entirely different units. However, this clarity comes with a significant tradeoff: percentile scoring is inherently relative to the specific study area. Consequently, the resulting values reflect comparative standings among peers rather than absolute levels of performance or achievement.

Federal Precedents and Update Cycles

Federal indices illustrate how tract- and neighborhood-level composite measures become institutionalized. The CDC/ATSDR Social Vulnerability Index (SVI), for example, aggregates multiple indicators into percentile-ranked theme scores and an overall score at the census tract level. The SVI is updated on a regular release cycle and is used operationally for disaster preparedness and response planning. The Area Deprivation Index (ADI) similarly measures neighborhood socioeconomic disadvantage using American Community Survey (ACS)-based indicators and is widely used in health services research and population risk stratification.

These precedents demonstrate that LEMI’s design aligns with a broader ecosystem of composite indices used to support public decision-making, particularly those relying on ACS 5-year estimates for small-area stability. Table 1 provides an overview of existing indicators alongside the LEMI.

Table 1: Comparison of Levers of Economic Mobility Index (LEMI), CDC Social Vulnerability Index (SVI), and Area Deprivation Index (ADI)

Feature	Levers of Economic Mobility Index (LEMI)	CDC Social Vulnerability Index (SVI)	Area Deprivation Index (ADI)
Primary Purpose	Measures structural conditions that shape economic mobility at the neighborhood level	Identifies communities that may need support before, during, or after disasters	Measures neighborhood socioeconomic disadvantage for health and policy analysis
Geographic Unit	Census Tract (2020 boundaries)	Census Tract	Census Block Group (often aggregated to tracts or counties)

Geographic Scope	Austin limited- and full-purpose boundaries	National (all U.S. tracts)	National (all U.S. block groups)
Core Domains	Health & Wellbeing; Livelihood & Work; Access & Belonging	Socioeconomic Status; Household Composition & Disability; Minority Status & Language; Housing & Transportation	Income; Education; Employment; Housing Quality
Data Sources	Primarily ACS 5-year estimates + supplemental datasets	ACS 5-year estimates	ACS 5-year estimates
Update Cycle	Periodic updates (Every two years)	Updated every two years using latest ACS 5-year estimates	Updated periodically following new ACS 5-year releases
Normalization Method	Percentile ranking within study area	Percentile ranking within state and national distributions	Factor analysis–based composite score; reported as state and national percentiles
Weighting Approach	Equal weighting across themes and sub-themes	Equal weighting across themes	Factor-derived weights based on principal components analysis
Output Scale	Percentile-ranked composite score (0–100) derived from averaged lever percentile ranks	Percentile ranks (0–100)	National and state percentile ranks (1–100)
Intended Use	Municipal planning, cross-sector analysis, economic mobility-focused decision support	Disaster preparedness, emergency response, public health targeting	Health research, healthcare planning, policy analysis
Interpretation	Relative structural conditions within the regional study area	Relative vulnerability compared to state and national distributions	Relative socioeconomic deprivation compared to state and national distributions

LEVER SELECTION

Selection Process and Criteria

Lever selection followed an iterative, structured process grounded in OECD guidance and municipal best practices. Candidate levers were reviewed for conceptual relevance to economic mobility, methodological reliability at the tract level, and usefulness for public decision-making.

The selection criteria emphasized relevance (capturing a structural condition tied to economic mobility), reliability (stable, well-documented data sources), actionability (use for planning and resource targeting), and non-redundancy (avoiding multiple measures of the same construct). Levers were also assessed for whether they reflect conditions that can change over time, supporting longitudinal monitoring without implying that neighborhood demographic composition is itself a “problem” to be solved.

Efforts to capture stakeholder and community input were also useful in informing lever selection. Those efforts are best described as a hybrid approach that captures historical community sentiment (Quality of Life study reviews), with community and committee meetings. For more information on community engagement efforts, please see the Austin Levers of Economic Mobility Index report.

Data Sources

Most levers are derived from American Community Survey (ACS) 5-year estimates because they provide the most reliable small-area coverage over time and are broadly replicable for future updates. Additional datasets were incorporated when they provided tract-level coverage aligned with the conceptual framework and could be processed consistently across the study region.

Included Levers

The LEMI includes 18 levers spanning the three themes. Levers were grouped into sub-themes to reflect the conceptual structure of the Index and to support interpretation by domain. Table 2 provides an overview of the levers included in the LEMI along with a description of the lever, data source, and product release cycle.

Table 2: Levers of Economic Mobility in Detail

Theme 1: Health & Wellbeing Sub-theme: Health Access and Resources				
Lever	Definition	Why This Lever?	Data Source	Data Product Release Cycle
Uninsured Rate	Percent of people (all ages) without health insurance.	Identifies gaps in access to preventive and emergency care that can undermine health stability and long-term economic mobility.	U.S. Census Bureau, American Community Survey, 2024 5-Year Estimates, Table S2701.	Annually
Enrollment in Early Education	Percent of 3- and 4-year-olds enrolled in early education programs (public and private).	Reflects the availability of foundational human capital development and early childhood support systems.	U.S. Census Bureau, American Community Survey, 2024 5-Year Estimates, Table S1401.	Annually
Life Expectancy	Estimated life expectancy of a census tract.	Reflects the cumulative impact of neighborhood conditions on long-term health, stability, and opportunity across the life course.	Every Texan analysis of CDC PLACES / USALEEP data crosswalked to 2020 tracts.	Periodically
Sub-theme: Wellbeing and Activity				
Disability Rate	Percent of people (all ages) who have at least one disability.	Identifies concentrations of residents with disabilities, signaling where accessibility, inclusive infrastructure, and supportive services are critical for full participation.	U.S. Census Bureau, American Community Survey, 2024 5-Year Estimates, Table S1810.	Annually
Seniors with Ambulatory Disabilities	Percent of people age 65+ who have an ambulatory disability.	Highlights mobility limitations among older adults that may restrict independence, service access, and community participation.	U.S. Census Bureau, American Community Survey, 2024 5-Year Estimates, Table S1810.	Annually

Low Physical Activity	Percent of adults 18+ with no leisure-time physical activity.	Reflects limited access to safe parks, sidewalks, and recreation facilities; signals opportunities to improve built-environment design.	CDC PLACES 2025 release, Behavioral Risk Factor Surveillance System (BRFSS) data from 2023.	Annually
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Theme 2: Livelihood & Work

Sub-theme: Income and Employment

Median Household Income	The midpoint where half of all households earn more and half earn less. The U.S. Census Bureau calculates this as the total pretax cash income of the householder and all other people age 15 and older in the home.	Measures local economic capacity and earning potential, reflecting access to stable employment and upward economic mobility.	U.S. Census Bureau, American Community Survey, 2024 5-Year Estimates, Table B19013.	Annually
Underemployment Rate	The percent of people age 16-64 (who worked in the past 12 months) who worked <35 hours per week and <27 weeks per year.	Identifies unstable or insufficient work patterns that limit income security and advancement.	U.S. Census Bureau, American Community Survey, 2024 5-Year Estimates, Table B23022.	Annually
Poverty Rate	Percent of people (all ages) living below the poverty threshold.	Captures direct economic hardship at the household level.	U.S. Census Bureau, American Community Survey, 2024 5-Year Estimates, Table S1701.	Annually

Sub-theme: Household Stability and Cost Burdens

Eviction Filing Rate	The number of eviction cases filed relative to the total number of renter-occupied households,	Signals housing instability and displacement risk that can disrupt employment,	2024 evictions filing data from Travis County courts. Compiled by BASTA (Building and	Annually
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	expressed as a percentage.	education, and community ties.	Strengthening Tenant Action).	
Household Support Risk Score	A composite measure that captures social vulnerability related to family structure by identifying areas where residents may face heightened caregiving burdens and limited household support, including concentrations of single parents with children, households with children, and elderly individuals living alone.	Identifies areas where caregiving demands and limited household support may strain financial stability and reduce resilience.	U.S. Census Bureau, American Community Survey, 2024 5-Year Estimates, Tables DP02 and B11007.	Annually
Energy Burden	Total annual housing energy costs divided by total annual household income, expressed as a percentage.	Reveals housing-related cost pressures that reduce disposable income and limit economic stability.	Ma, O., & Vimont, A. (2024). <i>Low-Income Energy Affordability Data - LEAD Tool - 2022 Update</i> . [Data set]. Open Energy Data Initiative (OEDI). U.S. Department of Energy, Office of Energy Efficiency and Renewable Energy. https://doi.org/10.25984/2504170	Periodically

Theme 3: Access & Belonging

Sub-theme: Education, Language, and Digital Access

Limited English Speaking Households	Percent of households that speak limited English.	Signals language-related access barriers that can limit engagement with services, civic participation, and economic opportunity.	U.S. Census Bureau, American Community Survey, 2024 5-Year Estimates, Table S1602.	Annually
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Households Without Internet	Percent of households with no internet.	Identifies digital access gaps that restrict educational attainment, workforce participation, and access to essential services.	U.S. Census Bureau, American Community Survey, 2024 5-Year Estimates, Table B28002.	Annually
Less Than High School Education	Percent of people 25+ whose highest educational attainment is less than a high school diploma.	Signals limited access to foundational credentials that shape employment stability, earnings potential, and long-term economic mobility.	U.S. Census Bureau, American Community Survey, 2024 5-Year Estimates, Table S1501.	Annually

Sub-theme: Environment and Neighborhood

Persistent Poverty Flag	Measure of whether an area is in concentrated poverty. The U.S. Census Bureau defines this as a census tract with a poverty rate of 20% or higher during the three decades period from 1989 to 2015-2019.	Identifies areas of concentrated and long-standing economic hardship that create structural barriers to upward economic mobility.	U.S. Census Bureau Persistent Poverty in Counties and Census Tracts, 1989 to 2015-2019.	Periodically. Determined by U.S. Census Bureau.
Institutional Group Quarters	Percent of individuals living in institutional group quarters. Institutional group quarters include correctional, juvenile, nursing, and other institutional facilities. Institutional group quarters do not include college or university housing, military quarters, or other noninstitutional facilities.	Highlights concentrations of residents in institutional or non-traditional housing settings where stability, autonomy, and access to opportunity may be constrained.	U.S. Census Bureau 2020 Decennial Redistricting Data (PL 94-171), Table P5.	Decennially

Heat Disparity	Difference between each 2020 census tract's maximum summer land surface temperature (June 1–August 31, 2024) and the study-area mean of tract-level summer maximum land surface temperature, measured in degrees Fahrenheit.	Identifies neighborhoods experiencing disproportionately higher summer heat exposure relative to the regional average. Elevated land surface temperatures are associated with reduced tree canopy, impervious surface concentration, higher household energy burden, and increased health risk. Heat exposure represents an environmental condition that can constrain economic mobility by affecting health stability, housing costs, and outdoor livability.	NASA Landsat 8 Land Surface Temperature (30m resolution) accessed via Climate Engine API; summarized to 2020 Census Tracts; Summer 2024 (June 1–August 31); maximum extremity calculation.	Seasonal (annual summer composite; updated as new Landsat data become available)
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Lever Construction Notes (Derived Levers)

Household Support Risk Score (HSRS)

HSRS is a tract-level composite lever designed to capture vulnerability related to caregiving burden and limited informal household support. It was constructed from ACS 5-year estimates using three components: (1) percent female-headed households with children, (2) percent households with children under 18, and (3) percent seniors (65+) living alone. For each component, tract-level percentages were calculated using standard ACS numerators and denominators (Tables DP02 and B11007). Each component was then converted to a percentile rank within the Austin limited- and full-purpose boundaries tract universe, and the HSRS was computed as the arithmetic mean of the three percentile-ranked component scores. Higher raw HSRS values represent higher support vulnerability; the lever was reversed during preprocessing to align with the Index convention that higher scores represent more favorable structural conditions.

Heat Disparity (Summer 2024 Maximum Land Surface Temperature Difference)

Heat Disparity was constructed as a relative exposure measure based on NASA Landsat 8 land surface temperature (LST) imagery at 30m resolution accessed via the Climate Engine API. The analysis window was meteorological summer (June 1–August 31, 2024). For each tract, mean LST was summarized for each Landsat overpass using an areal reducer, and a

“maximum extremity” value was then calculated as the maximum valid tract-level LST observed during the study window. Values were converted from Celsius to Fahrenheit. Heat Disparity was defined as the difference between each tract’s summer maximum LST and the study-area mean of summer maximum LST across all included tracts. Positive values indicate hotter-than-average tracts; negative values indicate cooler-than-average tracts. Because higher heat disparity reflects greater structural burden, the lever was reversed prior to percentile ranking to maintain consistent directionality across the Index.

Levers Considered but Not Included

During the development process, multiple potential levers were assessed but ultimately not incorporated into the final index. Exclusions were based on conceptual alignment, statistical performance at the census-tract level, and the potential for overlapping measurement of the same underlying construct.

Race and ethnicity were intentionally not included as scoring variables. The Index is designed to measure structural conditions, not demographic composition. Including racial or ethnic distribution as a scored component risks conflating population characteristics with neighborhood conditions and could create interpretive challenges if shifts in demographic makeup were mistaken for structural improvement or decline. Demographic patterns remain critically important for contextual analysis and are better examined through map overlays and stratified interpretation rather than embedded directly into the composite score.

Certain modeled health outcome datasets were also excluded where the underlying methodology relies heavily on socioeconomic inputs derived from the American Community Survey. Incorporating such measures would risk reintroducing the same socioeconomic variables already represented in the Index, effectively amplifying their influence through duplication rather than adding distinct explanatory value.

Transportation accessibility measures were reviewed but not included due to practical and methodological constraints. Reliable tract-level accessibility modeling requires detailed network analysis and frequent updating to remain accurate. Additionally, several transportation-related ACS variables exhibit limited statistical stability at small geographic scales. Given the goal of creating a maintainable and updateable index, these measures were not incorporated in the current version.

METHODOLOGY

Scoring Background

The LEMI is designed to support tract-to-tract comparability across a multi-county study region with diverse urban, suburban, and rural conditions. The Index therefore uses a normalization approach that places all levers onto a common, interpretable scale and reduces undue influence from extreme values. Percentile ranking was selected as the primary normalization method because it is robust to outliers and aligns with the approach used in widely adopted public indices. Percentile scores also translate well to mapping, where relative standing is often more relevant than raw units.

Directionality

Before normalization, all levers were reviewed for directionality so they aligned with the Index interpretation that higher scores indicate more favorable structural conditions. Most levers were originally oriented such that higher raw values reflected greater structural disadvantage; these were reversed during preprocessing. Enrollment in early education, life expectancy, and median household income were the three exceptions: because higher values already indicate better conditions, they retained their original orientation. As a result, all levers are oriented consistently, and index scores range from 0 to 100, with higher scores representing better-off census tracts.

Table 3: List of Levers Not Reversed

Lever	Direction
Enrollment in Early Education	Not Reversed
Life Expectancy	Not Reversed
Median Household Income	Not Reversed

Step-by-Step Computation

The LEMI score is derived through a multi-stage aggregation process. This approach ensures that levers with disparate units (e.g., currency, percentages, or raw counts) are comparable and weighted appropriately.

Phase 1: Data Pre-processing and Cleaning

The majority of levers included in the LEMI were obtained at the 2020 census tract level. Census tracts were included in the study area if at least 10% of the tract fell within Austin's limited-purpose or full-purpose boundaries. Tracts with less than 10% overlap were excluded from the analysis, leaving an initial study area of 250 tracts.

One lever required additional pre-processing to normalize to 2020 census tract boundaries. To mitigate statistical noise from non-residential areas (e.g., industrial zones or parks), tracts with a total population of less than 50 were excluded. One tract met this exclusion criterion and was removed from the final analysis, resulting in a final study area of 249 tracts.

Levers were inverted where necessary so that higher values consistently represent more favorable socioeconomic or environmental conditions. This phase also included the computation of two derived levers (HSRS and heat disparity) prior to normalization.

Phase 2: Normalization via Percentile Ranking

To allow for cross-lever comparison, raw lever values were converted to percentile ranks (PR) within the study area.

For each lever, a tract's percentile rank reflects its relative standing compared to all other tracts. A value of 80 indicates that the tract performs more favorably than approximately 80 percent of tracts on that lever.

This transformation standardizes all levers onto a common 0–100 scale while preserving relative differences.

Phase 3: Hierarchical Aggregation

Within each sub-theme and theme, the arithmetic mean of constituent lever percentile ranks was calculated. This additive aggregation method allows higher performance on one lever to offset lower performance on another within the same domain. All levers contribute equally to the composite, and the framework contains an equal number of levers within each theme and sub-theme.

- **Sub-theme score:** percentile-ranked mean of constituent lever percentiles
- **Theme score:** percentile-ranked mean of all lever percentiles within the theme
- **Overall composite score:** percentile-ranked mean of all lever percentiles across the Index

All final sub-theme, theme, and overall scores are expressed as the percentile rank of the mean of the constituent lever percentile ranks.

Importantly, the overall composite score is calculated directly from all lever percentile ranks rather than from theme-level or sub-theme-level scores. Theme and sub-theme scores therefore represent organizational summaries of the same additive lever framework and do not introduce hierarchical weighting in the overall index.

Phase 4: Final Percentile Ranking

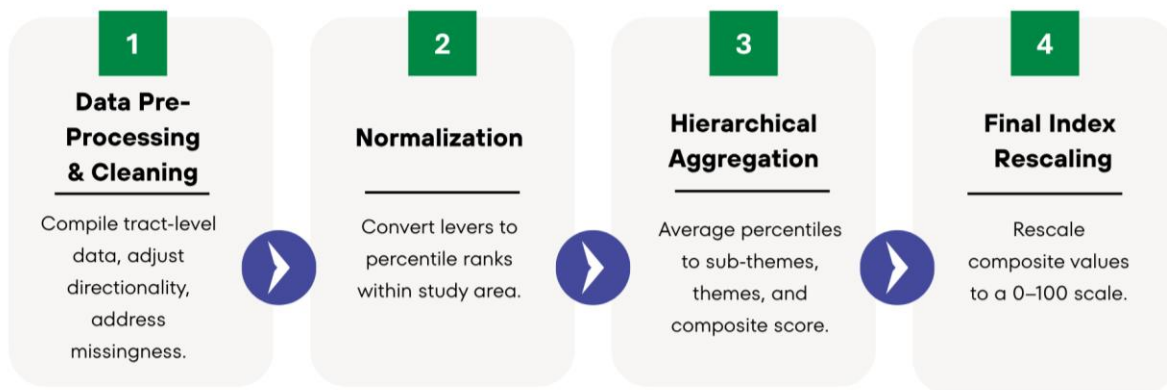
The overall composite score is interpreted as the final LEMI score within the study area. This step reflects each tract's relative standing based on the average of its lever-level percentile ranks.

Interpreting the LEMI

This final percentile rank constitutes the LEMI score as reported in the Index. A score of 15 indicates that the tract falls at approximately the 15th percentile relative to all other Austin census tracts, reflecting conditions less favorable than roughly 85% of tracts in the study area.

For more information on the calculation methods for the Calculate Composite Index tool, visit Esri's documentation [here](#).

Image A: LEMI Composite Index Construction Process



Weighting Structure

The LEMI applies equal weighting by design. Because each theme contains the same number of levers, calculating the overall composite score directly from all lever percentile ranks results in equal effective weighting across the three themes. Theme and sub-theme scores are provided for interpretation and do not introduce additional hierarchical weighting into the overall index. This weighting scheme reflects conceptual balance rather than allowing results to be driven by data volume or lever variance in any single domain.

This approach is consistent with OECD guidance that equal weighting is appropriate when strong theoretical rationale supports domain parity and when differential weighting cannot be justified by clear evidence or agreed-upon normative priorities.

Following OECD guidelines, a correlation threshold of 0.80 was established to identify potential double counting. Statistical analysis (Pearson's r) confirmed that no lever pairs exceeded this threshold. While some levers displayed moderate correlation (e.g., median household income and less than high school educational attainment), they were retained with Equal Weights as they represent conceptually distinct aspects of neighborhood vulnerability that should not be statistically penalized.

Data Harmonization and Imputation: Preprocessing

The LEMI dataset was developed through a two-stage process that first harmonized legacy geographic data and then utilized advanced statistical modeling to resolve remaining data gaps.

Geographic Crosswalking of Life Expectancy

Because the USALEEP Life Expectancy estimates were originally reported for 2010 census tract boundaries, areal-weighted interpolation was utilized to align these estimates with the 2020 geography used in the LEMI. This process allocated life expectancy values to 2020 tracts based on the specific land-area proportions where 2010 and 2020 boundaries overlapped. To ensure accuracy in areas with incomplete source data, a normalized life expectancy for each 2020 tract was calculated by dividing the weighted sum of values by the total weight of available data coverage. This methodology produced estimates aligned with the study area (defined as the Austin limited- and full-purpose boundaries tract universe) without performing imputation at this initial stage.

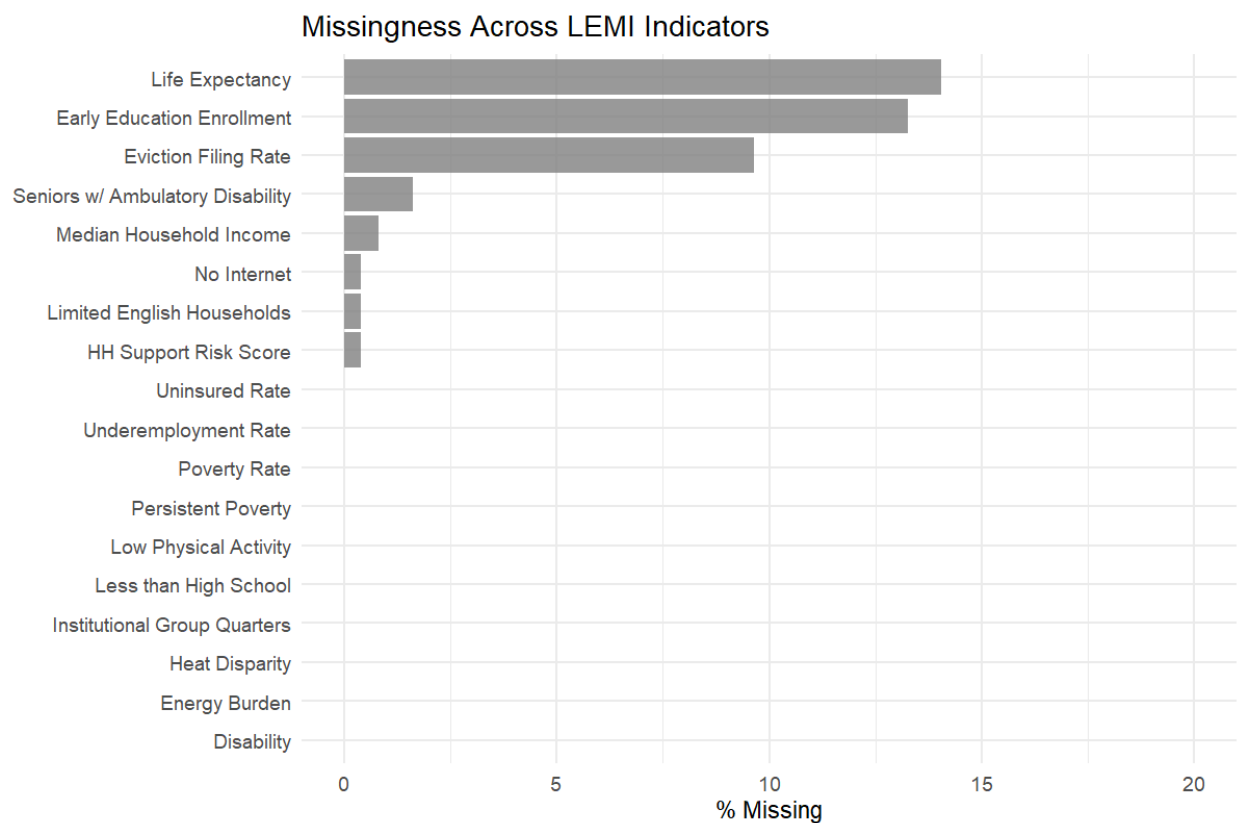
Multivariate Imputation by Chained Equations (MICE)

Following the geographic crosswalk, the workflow addressed missing values across 18 levers using Multivariate Imputation by Chained Equations (MICE) with a Predictive Mean Matching (PMM) approach. PMM was selected because it preserves the original distribution of the data by filling gaps with observed values from "donor" tracts that share similar characteristics. For each missing value, the algorithm identifies five potential donors and randomly selects one to ensure the imputed value is both realistic and within the existing data range.

The model was specifically constrained to prevent circularity and maintain the integrity of the lever relationships. For example, the script utilized a custom predictor matrix where eviction rates were imputed using only core socioeconomic predictors, while life expectancy was imputed using a combination of those core variables and health-specific levers like disability and physical activity levels.

Prior to imputation, 2.3% of values across the 18 levers were missing. Missingness was concentrated in a small number of variables rather than distributed across the dataset. Life expectancy had the highest rate of missing data at 14%, followed by enrollment in early education (13%), and eviction filing rate (10%). All remaining levers had 2% or fewer missing data. This pattern is consistent with data suppression in small-population census tracts rather than systematic measurement failure, supporting the appropriateness of multiple imputation as a remediation strategy. Figure 1 demonstrates the missingness pattern across the dataset.

Figure 1: Missingness Across LEMI Levers



To address these gaps, the MICE algorithm was run with 10 imputed datasets over 20 iterations, using Predictive Mean Matching with 5 donors per imputation. A critical assumption of MICE is that the iterative imputation process must converge; that is, the estimates produced by the algorithm must stabilize across iterations rather than trending in a particular direction. Figure 2 displays convergence diagnostics for the three levers with the highest missingness: life expectancy, enrollment in early education, and eviction filing rate. These variables also received custom predictor specifications and therefore required the closest scrutiny.

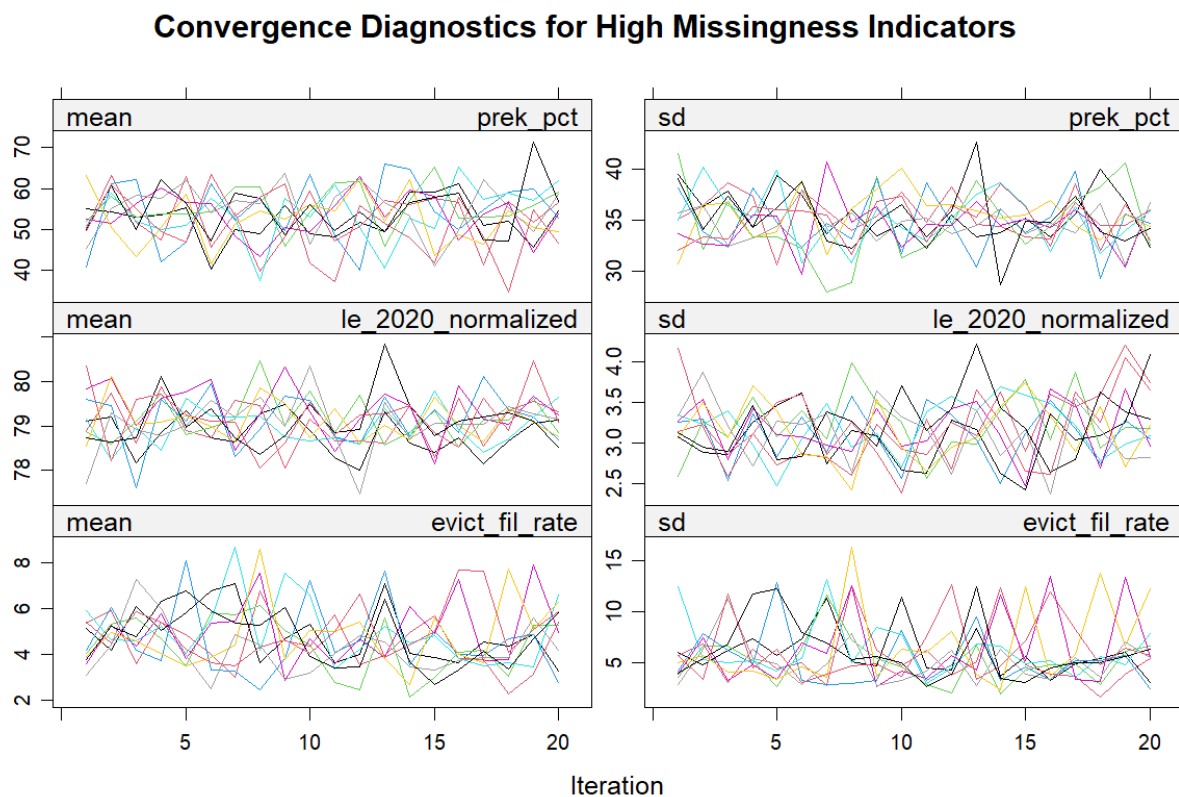


Figure 2: Convergence Diagnostics for High Missingness Levers

With convergence established, the next step was to verify that imputation preserved the underlying relationships among levers. If imputation had distorted these relationships, the resulting dataset would misrepresent the true structure of neighborhood disadvantage, undermining the validity of any composite index built from it. Figure 3 displays the pairwise correlation structure before and after imputation.

Visually, the two matrices are nearly identical, confirming that imputation did not distort the relationships among the levers. The overall similarity between the observed and post-imputation correlation matrices was 0.992, indicating near-perfect structural preservation. Strongly correlated variable pairs, such as poverty and underemployment, or median

household income and limited English-speaking households, maintained consistent relationships after imputation. No correlation pair shifted by more than 0.10 in absolute terms among substantively important variables with the exception of three pairs involving institutional group quarters: a variable with near-zero natural correlations where PMM introduced modest artificial structure through shared core predictors. This pattern is expected given the variable's distributional characteristics and does not meaningfully affect index construction.

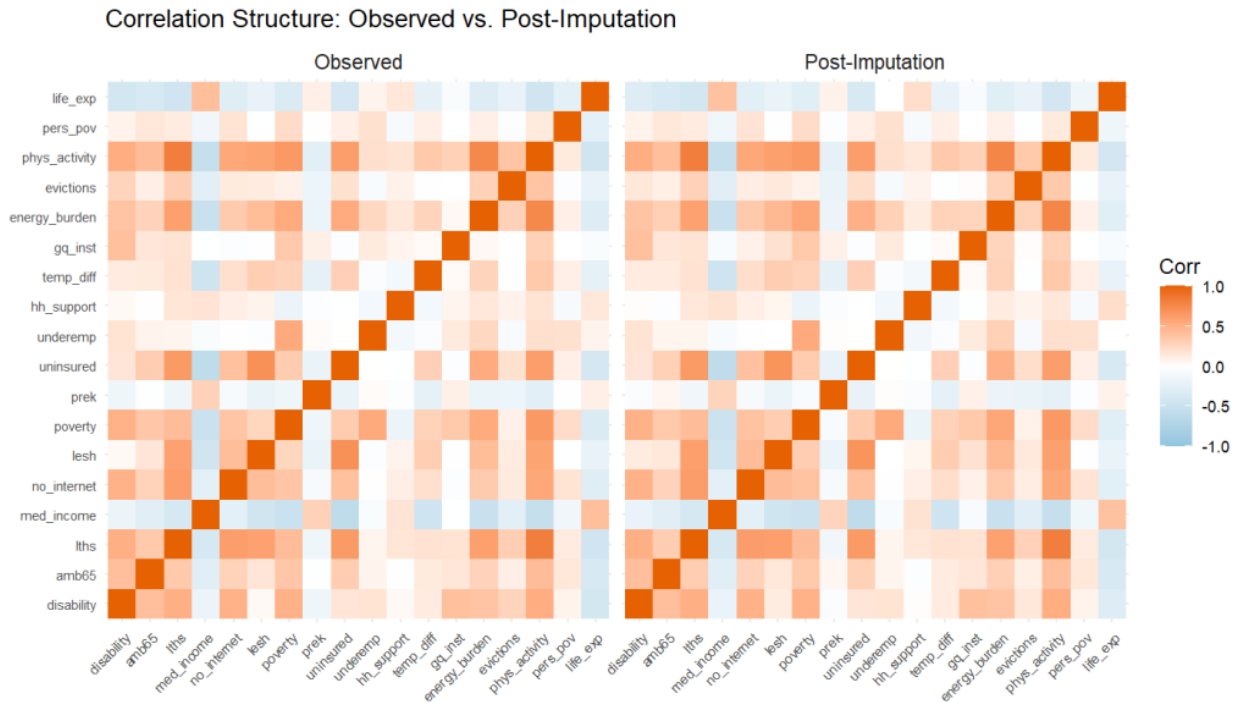


Figure 3: Correlation Structure

As a final check, Figure 4 examines whether imputed values for enrollment in early education (the sole ACS lever with missingness above 2%) fall within plausible ranges given socioeconomic context of each tract. Median household income was selected as the reference variable given its strong theoretical and empirical relationship with enrollment in early education rates (see references). This chart plots median household income against enrollment in early education rates for all census tracts, with imputed values shown in red and original observed values in grey. Imputed points are distributed throughout the same income range as observed points and follow the same general pattern, with no evidence of clustering at implausible extremes. This confirms that PMM drew from realistic donor pools rather than generating out-of-range synthetic values, and that the completed dataset is appropriate for use in index construction.

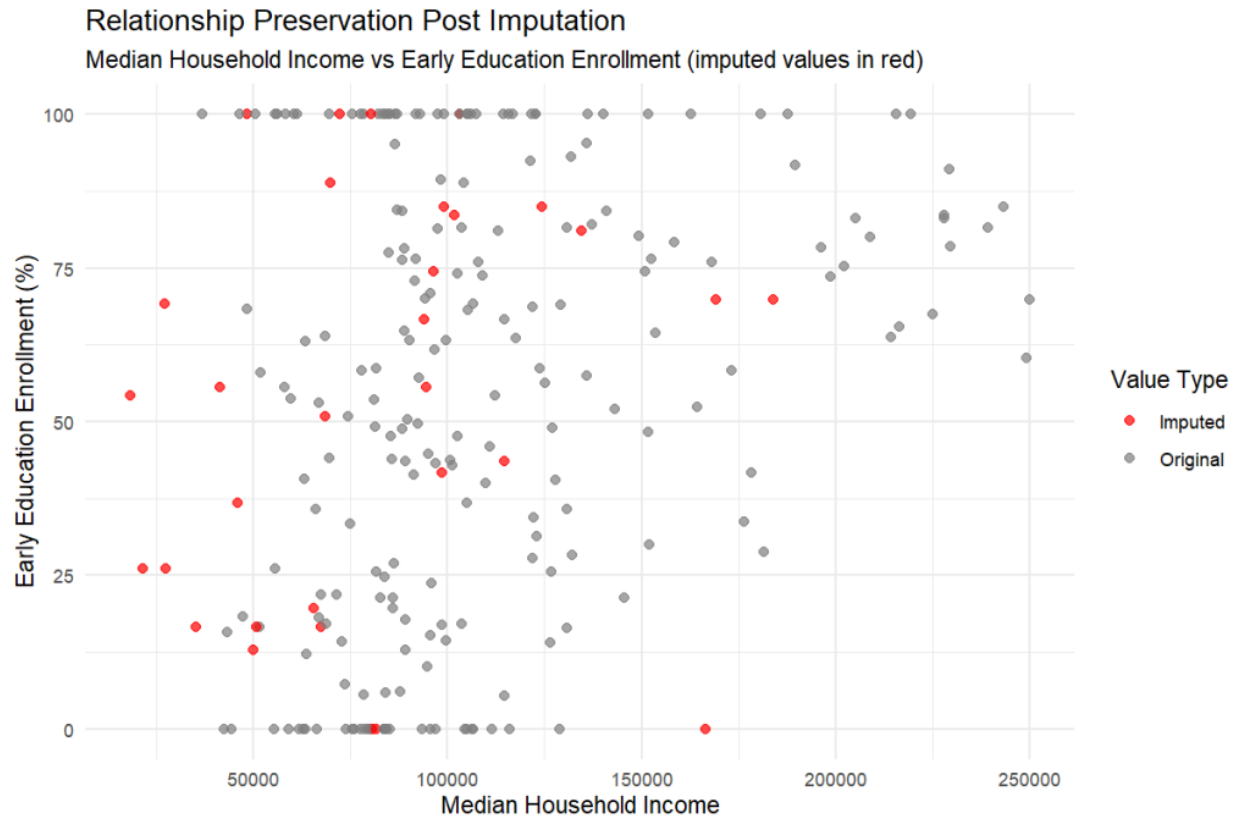


Figure 4: Example of Relationship Preservation Post Imputation

Summary of Validation and Integrity Checks for Imputation Methodology

The stability of the imputation process was verified through a series of rigorous diagnostics to ensure that the statistical properties of the dataset remained intact. The analysis confirmed that the correlation structure of the data was preserved, achieving a similarity metric of 0.992 between the observed and imputed correlation matrices. Furthermore, rank correlations remained near 1.00 for all levers, proving that the relative standing of census tracts was not distorted by the imputation process. HSRS and heat disparity were complete across the tract universe and did not require imputation.

Index Validation

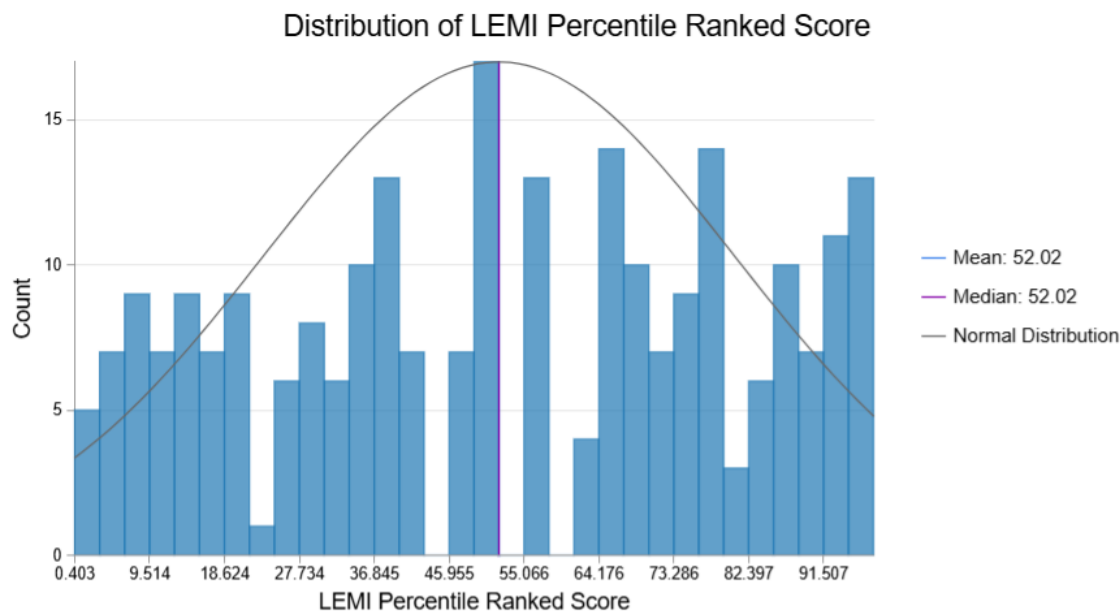


Figure 5: Distribution of LEMI Composite Index Levers Percentiles

The final spatial and statistical evaluation of the Levers of Economic Mobility Index (LEMI) across 249 census tracts reveals a robust and clear pattern of structural opportunity in Austin. The overall index exhibits a balanced and well-dispersed distribution with a mean of 52.02 and a median of 52.02. Because scores span nearly the full 0 to 100 range and lack significant ceiling or floor effects, the composite index meaningfully differentiates neighborhoods while maintaining internal stability. A primary step in this evaluation involved analyzing spatial patterns to identify general trends and outliers. A spatial hot spot analysis using the Getis–Ord Gi statistic confirmed significant clustering, with high-economic mobility tracts in western areas and southwestern areas and a statistical significance at the 99% confidence interval. A distinct cold spot cluster spans the eastern portion of the study area and extends into the southeastern periphery, with additional cold spot clustering emerging in the northeast. In these eastern and southeastern areas, multiple levers of economic mobility, such as underemployment and limited digital access, compound to create lower overall scores.

To ensure the 18 variables were balanced correctly, we utilized a box plot to compare the variance across the six sub-themes. This was essential because variables with higher variance, often caused by high internal correlation, can dominate the final index results. Our analysis confirmed that grouping levers into sub-indices and standardizing those sub-themes successfully equalized the variance. This methodological choice prevents the Livelihood & Work theme from overpowering other critical pillars like Health & Wellbeing or

Access & Belonging. We also addressed the binary persistent poverty lever by converting it to a percentile rank. This ensured the data was robust to the skewness inherent in binary variables and aligned its direction with the broader goal of improved economic mobility.

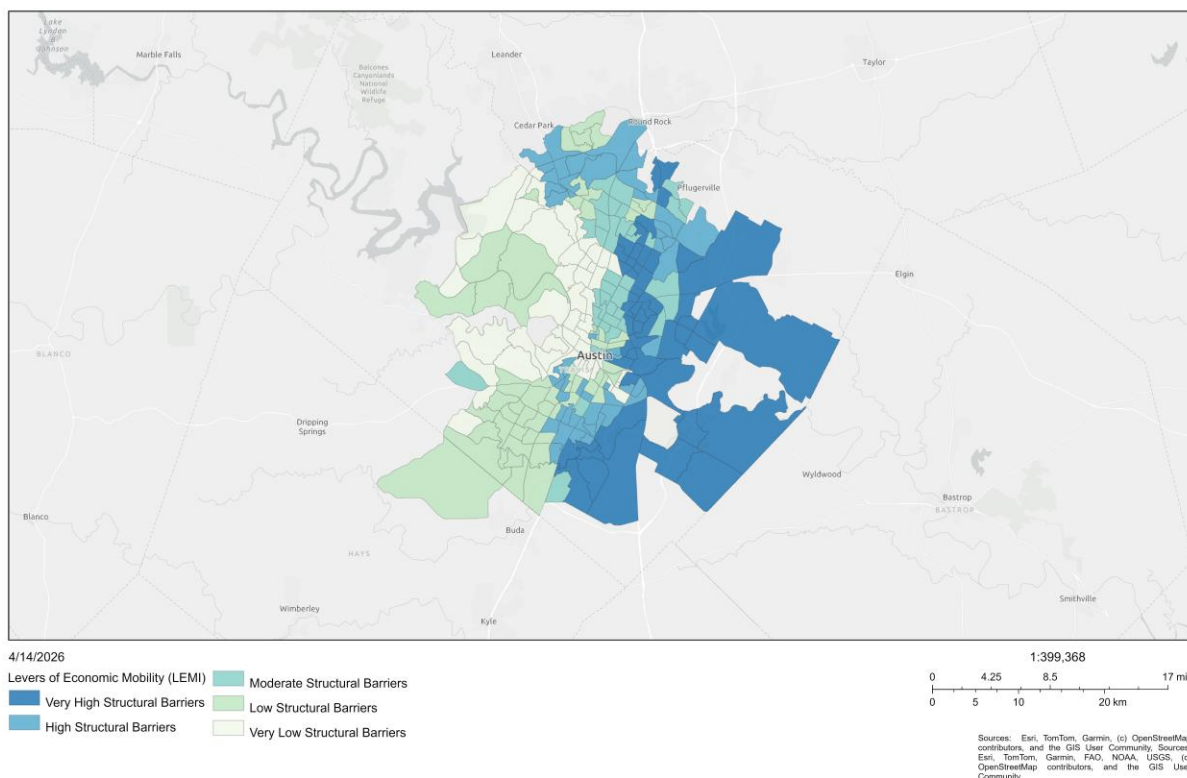
Finally, we conducted a sensitivity analysis to determine how much our methodological choices influenced the final results. By comparing the ranks produced from the "Mean of Percentiles" method against alternative combination methods, we found that tract rankings remained highly stable. This indicates that the LEMI results are robust and reflect underlying structural realities rather than being artifacts of the data processing steps.

Furthermore, the identification of spatial outliers, which are neighborhoods whose scores differ significantly from their immediate surroundings, helped us pinpoint specific tracts where local municipal interventions are successfully buffering residents against broader regional trends. These findings confirm that the Index captures geographically coherent patterns of opportunity and disadvantage rather than random spatial variation. Lastly, additional validation of the Index methodology was conducted in collaboration with a subject matter expert. Specifically, a subject matter expert reviewed and validated the preprocessing methodology.

RESULTS AND INTERPRETATION

The primary product of the LEMI is a tract-level map and associated theme maps that illustrate how structural conditions are distributed across the City and broader study area. The Index is intended to support place-based interpretation and to facilitate overlay analysis with policy-relevant geographies such as council districts, service areas, and capital planning zones.

Levers of Economic Mobility (LEMI)



LEMI scores should be interpreted as relative standings within the study region rather than as absolute measures. Tracts with percentile ranks closer to 0 reflect the least favorable structural conditions for economic mobility within the study area, while tracts near the 100th percentile reflect the most favorable conditions. Theme, sub-theme, and lever scores should be read the same way.

Theme scores and sub-theme scores provide interpretive clarity by identifying which dimensions contribute most to a tract's overall standing. The LEMI should be used as a diagnostic starting point that prompts deeper investigation rather than as a definitive label.

While all LEMI levers are normalized using percentile ranking for index construction, the levers persistent poverty and energy burden are presented differently in map tooltips due to their measurement structure. Persistent poverty is a binary lever identifying whether a tract meets the U.S. Census Bureau definition of long-term concentrated poverty (yes/no). Energy burden is reported in the source dataset as a small set of discrete percentage categories (0-4%), resulting in an ordinal scale with limited unique values.

Because persistent poverty and energy burden do not vary continuously across tracts, displaying them as percentile bars can imply artificial precision. In tooltip visualizations,

persistent poverty is therefore shown as a categorical status and energy burden is shown using ordinal category labels (none, low, moderate, high, or very high), while all other levers are displayed as percentile rankings. This visualization choice does not affect index computation and is intended solely to improve interpretability.

Spatial Patterns

The overall LEMI map reveals clear geographic patterning across the study area. Tracts exhibiting more favorable structural conditions are concentrated primarily in western and southwestern portions of the city, while tracts reflecting less favorable conditions are concentrated in eastern and southeastern areas. Central areas tend to display mixed conditions, indicating transitional or heterogeneous structural environments.

The observed west–east gradient suggests that economic mobility conditions are not randomly distributed but are spatially structured across neighborhoods. This pattern is consistent with broader socioeconomic stratification within the region and reflects the cumulative nature of housing markets, infrastructure investment, environmental conditions, and access to opportunity.

While the LEMI highlights spatial concentrations of relative advantage and disadvantage, it does not assign causality or characterize individual residents. Rather, it identifies structural context at the neighborhood level. Interpretation should therefore focus on place-based conditions and system-level factors rather than individual outcomes.

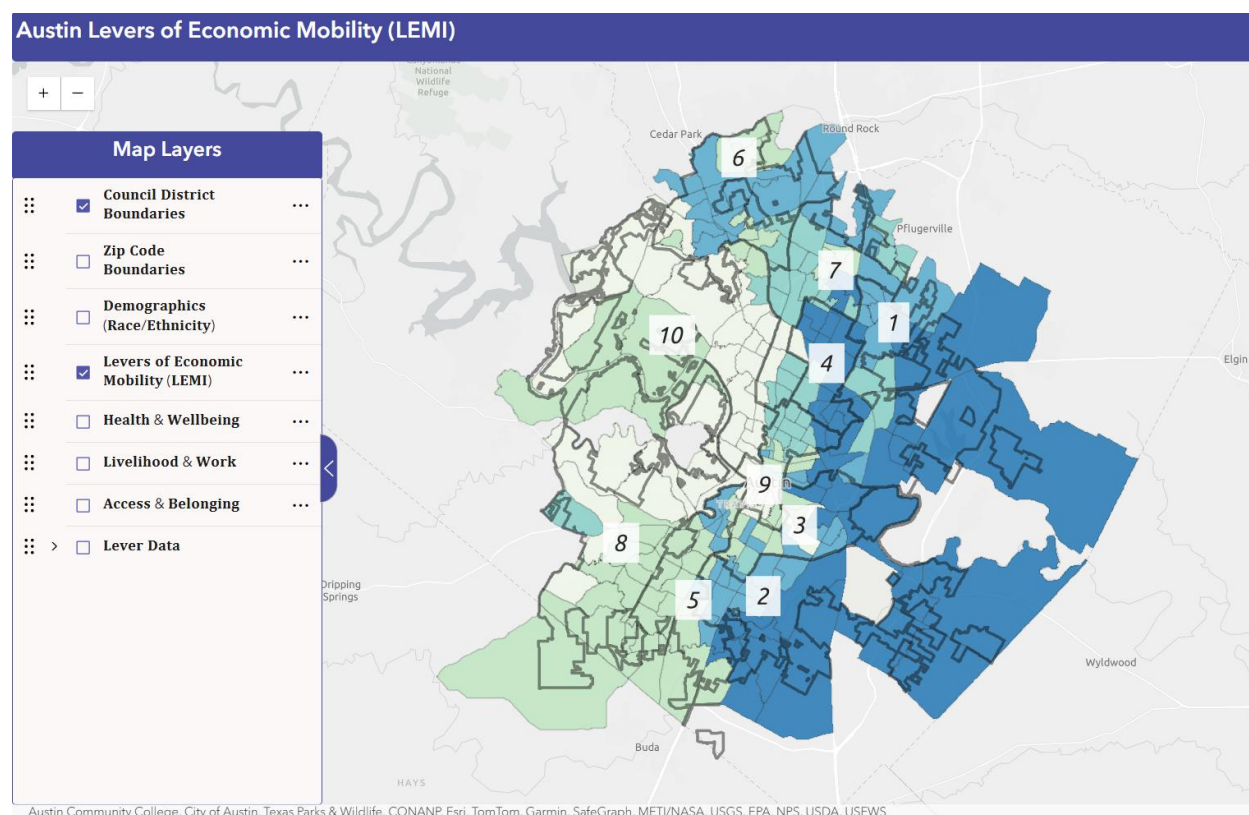
Using LEMI With Overlays

Within the LEMI ArcGIS experience application, multiple layers are available to explore alongside the Index itself. LEMI results can be paired with overlays such as City Council districts, zip codes, and demographic information. Additionally, users have the option to explore each of the individual themes (Health and Wellbeing, Livelihood and Work, and Access and Belonging). The interactive experience also allows users to bring in external datasets that are publicly available via ArcGIS online.

City Council District Overlay

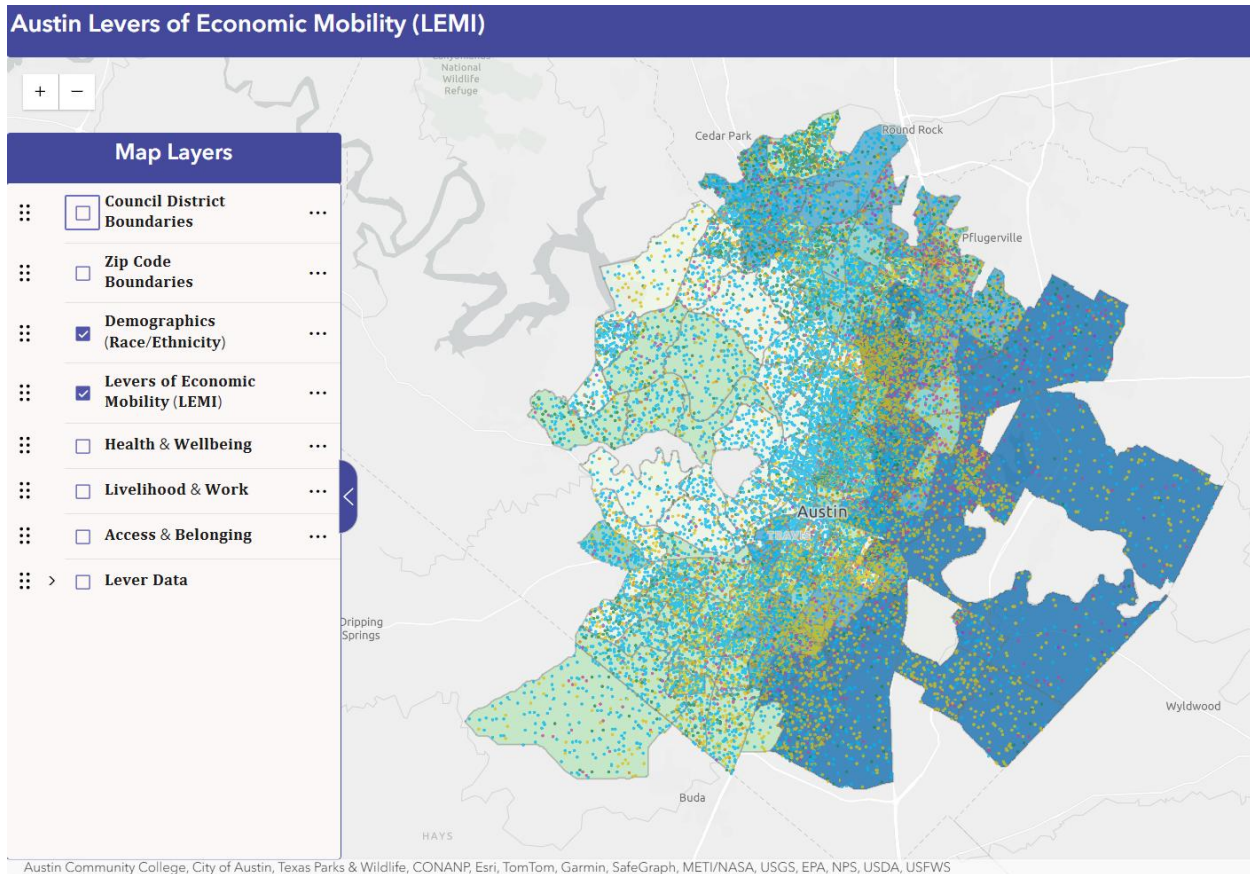
The inclusion of the City of Austin Council District boundary layer enhances the interpretability and policy relevance of the LEMI by linking tract-level structural conditions to council district geographies. While LEMI scores are calculated at the census tract level, overlaying council districts allows policymakers, staff, and community members to understand how patterns of economic mobility barriers align with district boundaries. This spatial alignment supports district-level planning, constituent engagement, and resource prioritization by enabling decision-makers to view concentrations of structural

disadvantage within their jurisdictions. The council district layer does not influence index calculation; rather, it serves as an interpretive tool that connects data-driven neighborhood conditions to governance and accountability structures.



Race/Ethnicity Overlay

The race and ethnicity overlay provides essential demographic context for interpreting LEMI patterns without incorporating race or ethnicity directly into the composite score. By displaying tract-level racial and ethnic composition alongside index results, users can examine how structural conditions intersect with population distribution while avoiding the methodological and ethical concerns of treating demographic composition itself as a scored deficit. This approach allows race and ethnicity to function as an analytic lens rather than an input variable, supporting transparent, context-informed interpretation of spatial disparities while maintaining the Index's focus on structural and socioeconomic conditions.



LIMITATIONS

The LEMI is designed to be transparent and maintainable, but it has limitations that should inform interpretation.

ACS 5-year estimates contain margins of error that vary by lever and geography, particularly at the tract level. Furthermore, because the Index relies on multi-year rolling averages, consecutive releases often contain overlapping data years; the LEMI should therefore be interpreted as a point-in-time diagnostic rather than a tool for longitudinal progress tracking. Additionally, percentile normalization produces relative rankings that may not reflect absolute change; a tract can improve in raw terms while maintaining a similar relative position. Finally, census tracts are proxies for neighborhoods and may mask meaningful within-tract variation.

The LEMI is best interpreted as a structured starting point for policy inquiry and geographic prioritization. It should be used alongside qualitative knowledge, community engagement, and sector-specific analyses.

CONCLUSION

The Levers of Economic Mobility Index (LEMI) provides a structured, transparent framework for assessing place-based conditions associated with economic mobility at the census tract level. By integrating multiple domains into a composite measure, the Index captures the cumulative structural environment in which households live, work, and access opportunity.

The LEMI is intended as a diagnostic tool rather than a prescriptive instrument. It highlights spatial patterns and relative standings that can inform strategic planning, resource allocation, and cross-sector collaboration. Used in conjunction with community input, administrative data, and policy context, the Index can help prioritize investments and guide deeper inquiry into neighborhood-level dynamics.

As with all composite indices, the LEMI reflects methodological choices and available data. It should be updated periodically to incorporate new data sources and evolving conditions. Future refinements may include additional levers, longitudinal analysis, or integration with outcome-based measures to further strengthen interpretive depth.

Ultimately, the value of the LEMI lies not in labeling places, but in supporting informed, equitable, and evidence-based decision-making across the city and broader region.

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